

Single-Score Safety Radius for the Damped Double Integrator

4D Lyapunov tube, exact 2D projection, and direct position
score

Tracking Problem

$$x(t) \in \mathbb{R}^4$$

Actual state
measured online

$$(x_d(t), u_d(t))$$

RRT expert
calibration only

$$(\hat{x}_d(t), \hat{u}_d(t))$$

LLM QP reference
available online

$$e(t) := x(t) - x_d(t), \quad \mathbb{P}[\|e(t)\|_2 \leq \rho(t), \forall t \in [0, T]] \geq 1 - \delta.$$

Calibration

$$A, B, K, P$$

$$\{(x_{d,i}, u_{d,i}, \hat{x}_{d,i}, \hat{u}_{d,i})\}_{i=1}^N$$

$$d_x := \hat{x}_d - x_d,$$

Online

$$x(t), \quad \hat{x}_d(t), \quad \hat{u}_d(t)$$

$$x_d(t), \quad u_d(t) \quad \text{unobserved}$$

$$d_u := \hat{u}_d - u_d.$$

Damped Double-Integrator Dynamics

State and design-level input:

$$x = [p_x \quad p_y \quad v_x \quad v_y]^T \in \mathbb{R}^4, \quad u = [u_x \quad u_y]^T \in \mathbb{R}^2.$$

With damping $d = 1.1$,

$$\dot{p} = v, \quad \dot{v} = -d v + d u, \quad \dot{x} = Ax + Bu,$$

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1.1 & 0 \\ 0 & 0 & 0 & -1.1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1.1 & 0 \\ 0 & 1.1 \end{bmatrix}.$$

The model assumes the low-level flight stack realizes the commanded planar input u .

Calibration QP: Exact ZOH Model

$$x_{k+1}^q = A_d x_k^q + B_d u_k^q, \quad q \in \{R, L\}.$$

$$a = e^{-d\Delta t}, \quad g = \frac{1-a}{d}, \quad h = \Delta t - g,$$

$$A_d = \begin{bmatrix} 1 & 0 & g & 0 \\ 0 & 1 & 0 & g \\ 0 & 0 & a & 0 \\ 0 & 0 & 0 & a \end{bmatrix}, \quad B_d = \begin{bmatrix} h & 0 \\ 0 & h \\ 1-a & 0 \\ 0 & 1-a \end{bmatrix}.$$

$$d = 1.1, \quad \Delta t = 0.1 \text{ s}.$$

Calibration QP: Shared Clock and Knots

$$r_j^q = [p_{x,j}^q \quad p_{y,j}^q]^\top, \quad q \in \{R, L\}.$$

$$T_j^q = \max \left\{ \frac{\|r_{j+1}^q - r_j^q\|_2}{v_c}, 0.5 \right\}, \quad v_c = 0.30 \text{ m/s}.$$

$$H = \max \left\{ m_R, \text{round} \left(\frac{\sum_{j=0}^{m_R-1} T_j^R}{\Delta t} \right) \right\}, \quad \sum_{j=0}^{m_q-1} n_j^q = H.$$

$$\kappa_0^q = 0, \quad \kappa_j^q = \sum_{\ell=0}^{j-1} n_\ell^q, \quad t_k = k\Delta t, \quad k = 0, \dots, H.$$

Calibration QP: Minimum-Control Solution

$$\begin{aligned} \min_{u_0^q, \dots, u_{H-1}^q} \quad & \frac{1}{2} \sum_{k=0}^{H-1} \|u_k^q\|_2^2 \\ \text{s.t.} \quad & x_{k+1}^q = A_d x_k^q + B_d u_k^q, \quad k = 0, \dots, H-1, \\ & x_0^q = [(r_0^q)^\top \quad 0 \quad 0]^\top, \\ & [I_2 \quad 0] x_{\kappa_j^q}^q = r_j^q, \quad j = 1, \dots, m_q, \\ & [0 \quad I_2] x_H^q = 0. \\ & U^q = [(u_0^q)^\top \quad \dots \quad (u_{H-1}^q)^\top]^\top. \\ \min_{U^q} \quad & \frac{1}{2} \|U^q\|_2^2 \quad \text{s.t.} \quad C^q U^q = b^q. \end{aligned}$$

$$(U^q)^* = (C^q)^\top (C^q (C^q)^\top)^{-1} b^q.$$

Implementable Control Law

$$\dot{x}_d = Ax_d + Bu_d, \quad \dot{\hat{x}}_d = A\hat{x}_d + B\hat{u}_d.$$

Define true tracking error and prediction gaps:

$$e := x - x_d, \quad d_x := \hat{x}_d - x_d, \quad d_u := \hat{u}_d - u_d.$$

The implementable controller tracks the available LLM reference:

$$\hat{e} := x - \hat{x}_d = e - d_x, \quad u = -K\hat{e} + \hat{u}_d = -K(e - d_x) + \hat{u}_d.$$

$$\underbrace{x_d, u_d}_{\text{calibration only}}, \quad \underbrace{x, \hat{x}_d, \hat{u}_d}_{\text{online}}.$$

Error Dynamics: Combine the Two Gaps First

Starting with $e = x - x_d$,

$$\begin{aligned}\dot{e} &= Ax + Bu - (Ax_d + Bu_d) \\ &= Ae + B[-K(e - d_x) + \hat{u}_d - u_d] \\ &= (A - BK)e + Bd_u + BKd_x.\end{aligned}$$

Define

$$A_c := A - BK, \quad \boxed{w(t) := Bd_u(t) + BKd_x(t)}.$$

Then the entire uncertain error system is

$$\boxed{\dot{e} = A_c e + w.}$$

The closed loop sees only w ; it does not see separate worst-case values of d_u and d_x .

Lyapunov Metric and Certified Decay Rate

Choose $P = P^\top \succ 0$ and

$$V(e) := e^\top P e.$$

The required certificate condition is

$$A_c^\top P + P A_c \preceq -2\alpha_P P, \quad \alpha_P > 0.$$

For fixed P, K , the largest valid rate is

$$\alpha_P = \frac{1}{2} \lambda_{\min} \left(P^{-1/2} \left[-A_c^\top P - P A_c \right] P^{-1/2} \right).$$

Do not substitute the spectral abscissa automatically

– $\max \operatorname{Re} \lambda(A_c)$ describes modal decay, but it need not satisfy the matrix inequality in the chosen P -metric.

Differentiate the Lyapunov Function

Using $\dot{e} = A_c e + w$,

$$\begin{aligned}\dot{V} &= \dot{e}^\top P e + e^\top P \dot{e} \\ &= e^\top (A_c^\top P + P A_c) e + 2e^\top P w \\ &\leq -2\alpha_P V + 2e^\top P w.\end{aligned}$$

Factor $P = P^{1/2} P^{1/2}$ and apply Cauchy–Schwarz:

$$\begin{aligned}e^\top P w &= (P^{1/2} e)^\top (P^{1/2} w) \\ &\leq \|P^{1/2} e\|_2 \|P^{1/2} w\|_2 \\ &= \sqrt{V} \sqrt{w^\top P w}.\end{aligned}$$

Therefore

$$\dot{V} \leq -2\alpha_P V + 2\sqrt{V} \sqrt{w^\top P w}.$$

Scalar Comparison System

Let

$$W := \sqrt{V} = \sqrt{e^\top P e}.$$

For $W > 0$, $\dot{W} = \dot{V}/(2W)$, hence

$$\dot{W} \leq -\alpha_P W + \sqrt{w^\top P w}.$$

If a trajectory satisfies

$$\sup_{t \in [0, T]} \sqrt{w(t)^\top P w(t)} \leq q_w,$$

then

$$\dot{W} \leq -\alpha_P W + q_w.$$

Solving this scalar differential inequality gives

$$W(t) \leq W(0)e^{-\alpha_P t} + \frac{q_w}{\alpha_P}(1 - e^{-\alpha_P t}).$$

One Conformal Nonconformity Score

For calibration environment i , compute the full trajectories and form

$$w_i(t) = B(\hat{u}_{d,i}(t) - u_{d,i}(t)) + BK(\hat{x}_{d,i}(t) - x_{d,i}(t)).$$

Use one scalar score:

$$S_i := \max_{t \in [0, T_i]} \sqrt{w_i(t)^\top P w_i(t)}.$$

Given N exchangeable calibration scores, set

$$k = \lceil (1 - \delta)(N + 1) \rceil, \quad q_w = S_{(\min\{k, N\})},$$

where $S_{(j)}$ is the j th order statistic.

Then

$$\mathbb{P}(S_{\text{test}} \leq q_w) \geq 1 - \delta.$$

Because S_i already maximizes over time, this is a simultaneous trajectory-level event.

Convert the Metric Bound to a State-Norm Tube

Eigenvalue bounds give

$$\sqrt{\lambda_{\min}(P)} \|e\|_2 \leq \sqrt{e^\top P e} = W \leq \sqrt{\lambda_{\max}(P)} \|e\|_2.$$

Thus

$$\rho(t) = \sqrt{\frac{\lambda_{\max}(P)}{\lambda_{\min}(P)}} \|e(0)\|_2 e^{-\alpha_P t} + \frac{q_w}{\alpha_P \sqrt{\lambda_{\min}(P)}} (1 - e^{-\alpha_P t}).$$

With probability at least $1 - \delta$,

$$\|x(t) - x_d(t)\|_2 \leq \rho(t), \quad \forall t \in [0, T].$$

As $t \rightarrow \infty$,

$$\rho_\infty = \frac{q_w}{\alpha_P \sqrt{\lambda_{\min}(P)}}.$$

Numerical CARE Solution Used in the Repository

The current LQR weights are

$$Q = \text{diag}(10, 10, 1, 1), \quad R = I_2.$$

The CARE solution is

$$P = \begin{bmatrix} 8.8032 & 0 & 2.8748 & 0 \\ 0 & 8.8032 & 0 & 2.8748 \\ 2.8748 & 0 & 1.6216 & 0 \\ 0 & 2.8748 & 0 & 1.6216 \end{bmatrix},$$

$$K = \begin{bmatrix} 3.1623 & 0 & 1.7838 & 0 \\ 0 & 3.1623 & 0 & 1.7838 \end{bmatrix}.$$

$$\lambda_{\min}(P) = 0.612622, \quad \lambda_{\max}(P) = 9.812202,$$

$$\sqrt{\kappa(P)} = 4.002092.$$

Certified Rate Versus Eigenvalue Rate

Closed-loop eigenvalues:

$$\lambda(A - BK) = -1.531095 \pm 1.065013j \quad (\text{each repeated twice}).$$

The code currently reports the spectral value

$$\alpha_{\text{eig}} = 1.531095.$$

But the generalized-eigenvalue calculation required by the P -metric gives

$$\alpha_P = 1.096230.$$

Indeed,

$$A_c^\top P + PA_c + 2\alpha_{\text{eig}}P \not\preceq 0,$$

whereas

$$A_c^\top P + PA_c + 2\alpha_P P \preceq 0.$$

The radius calculation must use α_P unless a different valid transient bound is proved.

2,000-Sample Single-Score Calibration

Calibration source:

- $N = 2000$ actual refined and verified RRT/LLM prediction pairs;
- shared-clock minimum-control QP trajectories;
- score $S_i = \max_t \sqrt{w_i^\top P w_i}$;
- desired trajectory-level coverage $1 - \delta = 0.90$.

Finite-sample rank:

$$k = \lceil 0.9(2000 + 1) \rceil = \lceil 1800.9 \rceil = 1801.$$

The empirical conformal quantile is

$$q_w = S_{(1801)} = 1.641463.$$

Only one failure budget $\delta = 0.10$ is used; no $\delta_u + \delta_x$ split is required.

Numerical Radius

Insert

$$\begin{aligned}q_w &= 1.641463, & \alpha_P &= 1.096230, \\ \lambda_{\min}(P) &= 0.612622, & \sqrt{\kappa(P)} &= 4.002092.\end{aligned}$$

Steady-state term:

$$\begin{aligned}\rho_\infty &= \frac{q_w}{\alpha_P \sqrt{\lambda_{\min}(P)}} \\ &= \frac{1.641463}{1.096230 \sqrt{0.612622}} \\ &= \boxed{1.913079}.\end{aligned}$$

Complete time-dependent tube:

$$\begin{aligned}\rho(t) &= 4.002092 \|e(0)\|_2 e^{-1.096230t} \\ &\quad + 1.913079(1 - e^{-1.096230t}).\end{aligned}$$

If $e(0) = 0$, the tube grows monotonically from zero toward 1.913079.

Projecting the 4D Ellipsoid onto 2D Position

Let the position output be

$$e_p = C_p e, \quad C_p = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

Suppose the 4D Lyapunov argument has established

$$e^\top P e \leq \beta(t)^2.$$

The smallest Euclidean position ball containing the projection is found from

$$\rho_p(t)^2 = \max_{e^\top P e \leq \beta(t)^2} e^\top C_p^\top C_p e.$$

Set $z = P^{1/2}e$. The constrained maximization gives

$$\rho_p(t) = g_p \beta(t), \quad g_p := \sqrt{\lambda_{\max}(C_p P^{-1} C_p^\top)}.$$

Why the full inverse appears

Position and velocity are coupled by the off-diagonal blocks of P . Using only the upper-left position block of P is therefore not exact.

Projected Position Tube from the 4D Certificate

The scalar Lyapunov bound is

$$\beta(t) = W(0)e^{-\alpha_P t} + \frac{q_w}{\alpha_P}(1 - e^{-\alpha_P t}), \quad W(0) = \sqrt{e(0)^\top P e(0)}.$$

Therefore the inherited 2D output certificate is

$$\rho_p(t) = g_p W(0)e^{-\alpha_P t} + g_p \frac{q_w}{\alpha_P}(1 - e^{-\alpha_P t}).$$

For the repository CARE matrix,

$$C_p P^{-1} C_p^\top = 0.269772 \, I_2, \quad g_p = 0.519396.$$

For the original $q_w = 1.641463$ calibration and $e(0) = 0$,

$$\rho_{p,\infty} = 0.519396 \frac{1.641463}{1.096230} = 0.777728 \text{ m.}$$

This is tighter than 1.913079, while inheriting the same 4D conformal event.

0.30 m/s: Projection Versus Direct Score

Reducing the QP cruise speed and recalibrating the old 4D disturbance score gives

$$q_w = 1.271204, \quad \rho_{4D,\infty} = 1.481553 \text{ m},$$
$$\rho_{p,\infty}^{\text{proj}} = 0.602299 \text{ m}.$$

The deployed pipeline instead calibrates the position output directly. For closed-loop model trajectory $p_i^{\text{cl}}(t_k)$ and expert path $\gamma_{d,i}$,

$$S_i^{\text{cross}} = \max_k \min_s \left\| p_i^{\text{cl}}(t_k) - \gamma_{d,i}(s) \right\|_2.$$

At $N = 2000$, $\delta_p = 0.10$, and rank 1801,

$$q_p = S_{(1801)}^{\text{cross}} = 0.241187 \text{ m}, \quad \rho_p = q_p.$$

No P , α_P , or eigenvalue conversion is applied to this direct score.

What Each Position Number Guarantees

Construction	Radius	Contract
Original 4D norm	1.913079 m	$\ [e_p, e_v] \ _2$
2D projection of original 4D tube	0.777728 m	$\ e_p \ _2$ at equal time
0.30 m/s projected 4D tube	0.602299 m	equal-time position
Direct equal-time position score	0.255089 m	sampled $\max_k \ e_p(t_k) \ _2$
Direct cross-track position score	0.241187 m	sampled path distance

- Projection protects against every position–velocity combination inside the 4D ellipsoid.
- Direct calibration scores the output instead of enclosing the full state.
- Cross-track does not bound along-track timing error or velocity error. The current calculation uses exact QP zero-order hold and 20 Hz samples.

Why One Score Is Tighter

The previous two-score construction first applies

$$\|Bd_u + BKd_x\| \leq \|B\|\|d_u\| + \|BK\|\|d_x\|,$$

then combines two separate conformal events with a union bound.

The one-score construction calibrates

$$\sqrt{(Bd_u + BKd_x)^\top P(Bd_u + BKd_x)}$$

directly. It retains:

- the directions of d_u and d_x ;
- their correlation within each trajectory;
- cancellation or reinforcement through B and BK ;
- the exact certificate metric P .

The smaller radius comes from removing norm and union-bound conservatism, not from weakening 90% coverage.

What Is Fixed and What Must Be Recalibrated

Fixed for one certificate

- dynamics matrices A, B ;
- controller gain K ;
- metric P ;
- certified rate α_P ;
- score definition and coverage $1 - \delta$.

Requires recalibration

- changing K or LQR Q, R ;
- changing P or state normalization;
- changing trajectory generation;
- changing clock alignment;
- changing the operating distribution.

Reason

The score contains both P and K through $\sqrt{w^\top P w}$ and $w = B d_u + B K d_x$.

Interpretation and Assumptions

- The conformal guarantee requires calibration and test environments to be exchangeable.
- The linear model and idealized low-level realization must remain valid over the certified operation.
- The state norm mixes position and velocity components; 1.913079 is a full-state Euclidean bound, not automatically a pure geometric distance.
- Since $\|p - p_d\|_2 \leq \|x - x_d\|_2$, it is numerically conservative for position. A unit-consistent position certificate should use explicit state normalization or output projection.
- This bounds RRT-LLM discrepancy entering the error dynamics; it does not certify obstacle clearance by itself.

Final 90% single-score certificate

$$\mathbb{P}[\|x(t) - x_d(t)\|_2 \leq 4.002092\|e(0)\|e^{-1.096230t} + 1.913079(1 - e^{-1.096230t}), \forall t] \geq 0.90.$$

For the current direct-output deployment, the separate calibrated statement is $\mathbb{P}[S_{\text{test}}^{\text{cross}} \leq 0.241187 \text{ m}] \geq 0.90$; it must not be described as a 4D state bound.